Angle Grinder



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Angle Grinder

What it is used for

An angle grinder uses a rotating abrasive or cutting disc to grind, cut, or finish suitable materials. Select the correct disc and speed rating, secure the workpiece, and use the guard and appropriate PPE to reduce the risk of disc failure, sparks, and serious injury.

Main parts

- Grinder body and electric motor
- Spindle and disc mounting flange
- Compatible grinding or cutting disc
- Adjustable protective guard
- Auxiliary side handle
- Power switch, cord, and spindle lock

Before you start

1. Confirm that the grinder and disc are suitable for the material and intended operation.
2. Inspect the grinder, power cord, plug, guard, handle, and disc for damage or wear.
3. Check the disc for cracks, chips, moisture damage, and its maximum speed rating.
4. Ensure the disc's rated speed is at least the grinder's no-load speed and that the disc fits correctly.
5. Secure the workpiece in a vice or suitable fixture; remove flammable materials from the spark path.
6. Wear safety glasses and a face shield, hearing protection, suitable gloves, and appropriate protective clothing; use respiratory protection when required.

How to use

1. Disconnect the grinder from power before fitting or changing a disc.
2. Install the correct disc and flange in the orientation specified by the manufacturer.
3. Tighten the mounting nut using the supplied tool without excessive force.
4. Fit and position the guard between the disc and the operator, and attach the auxiliary handle.
5. Check that the disc rotates freely by hand while the tool is disconnected.
6. Connect power and hold the grinder firmly with both hands using the intended grip.
7. Position yourself so that sparks and debris are directed away from people and flammable materials.
8. Start the grinder and allow it to reach operating speed before contacting the workpiece.
9. Apply light, controlled pressure and keep the disc at the angle recommended for the task.

10. Use the correct disc face for grinding; do not side-load a cutting disc.
11. Avoid twisting, jamming, or forcing the disc through the material.
12. Release the switch and wait for the disc to stop completely before setting the tool down.
13. Disconnect power before inspecting the disc, clearing debris, or changing accessories.
14. After the tool cools, clean it and inspect the disc and guard before storage.

⚠ Safety

- Never operate the grinder without the correct guard and auxiliary handle where designed for use.
- Do not use damaged, incorrectly rated, or incompatible discs.
- Keep hands, cables, clothing, and body parts away from the rotating disc.
- Beware of sparks, hot metal, kickback, and flying fragments; keep bystanders clear.
- Never force a cutting disc sideways or use a cutting disc for side grinding.
- Disconnect power before adjustments, accessory changes, or maintenance; follow lockout procedures where applicable.

Care & storage

- Clean ventilation openings and the tool exterior after use, with power disconnected.
- Inspect the guard, handle, spindle, cord, and switch regularly.
- Replace worn or damaged discs and accessories; store discs dry and protected from impact.
- Arrange servicing if the grinder vibrates excessively, sparks internally, or makes unusual noises.
- Store the grinder securely with its cord protected from cuts and crushing.

Common mistakes to avoid

- Removing the guard or holding the grinder with only one hand.
- Using a disc with an unsuitable speed rating or visible damage.
- Applying excessive pressure or twisting the disc during a cut.
- Setting the grinder down before the disc stops completely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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